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Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding conditional UI.

Create:

feature/conditional-ui

Screenshot:

Terminal showing your current branch

```
WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git checkout -b feature/conditional-ui
Switched to a new branch 'feature/conditional-ui'

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)
$ ggit branch
bash: ggit: command not found

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)
$ git branch
* feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
  feature/student-interactions
  feature/student-list
  feature/widgets
  main
```

🚩 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an **active** status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

Active

Inactive

Test

Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

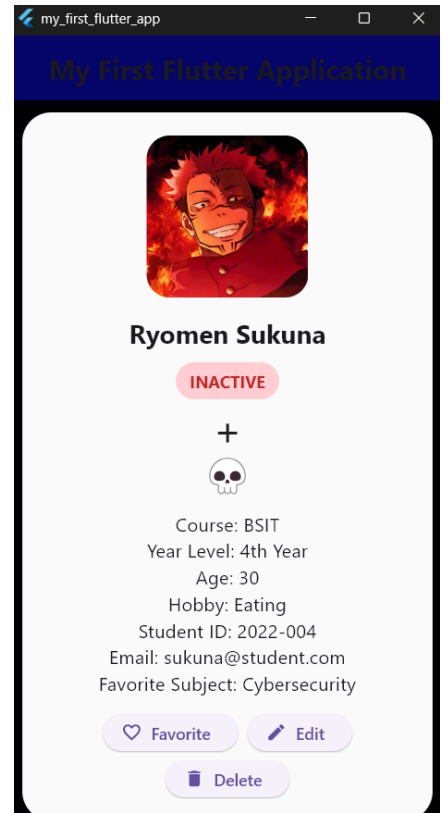
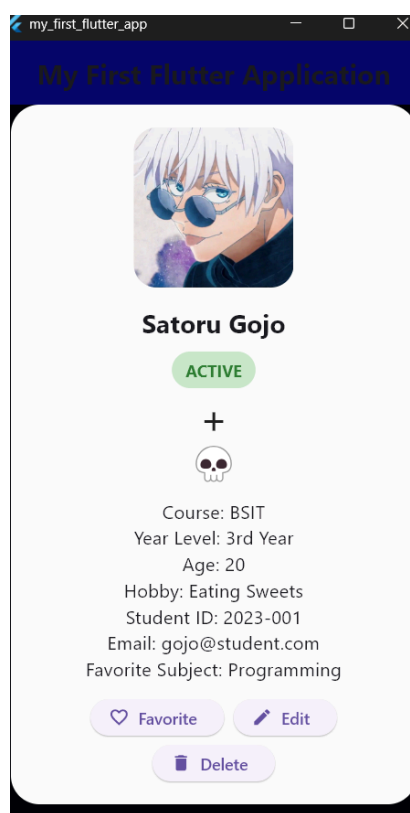
Screenshot:

```
final String? favorite;  
final bool? active;
```

```
favoriteSubject  
active: true,
```

```
active: false,
```

```
profile.active == true  
? Container(  
  padding: const EdgeInsets.symmetric(  
    horizontal: 12,  
    vertical: 6,  
  ), // EdgeInsets.symmetric  
  decoration: BoxDecoration(  
    color: Colors.green.shade100,  
    borderRadius: BorderRadius.circular(20),  
  ), // BoxDecoration  
  child: Text(  
    "ACTIVE",  
    style: TextStyle(  
      color: Colors.green.shade800,  
      fontWeight: FontWeight.bold,  
    ), // TextStyle  
  ), // Text  
) // Container  
: Container(  
  padding: const EdgeInsets.symmetric(  
    horizontal: 12,  
    vertical: 6,  
  ), // EdgeInsets.symmetric
```



🚩 Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

Favorite badge appears only when favorited

Warning appears only when inactive

Delete action appears only if active

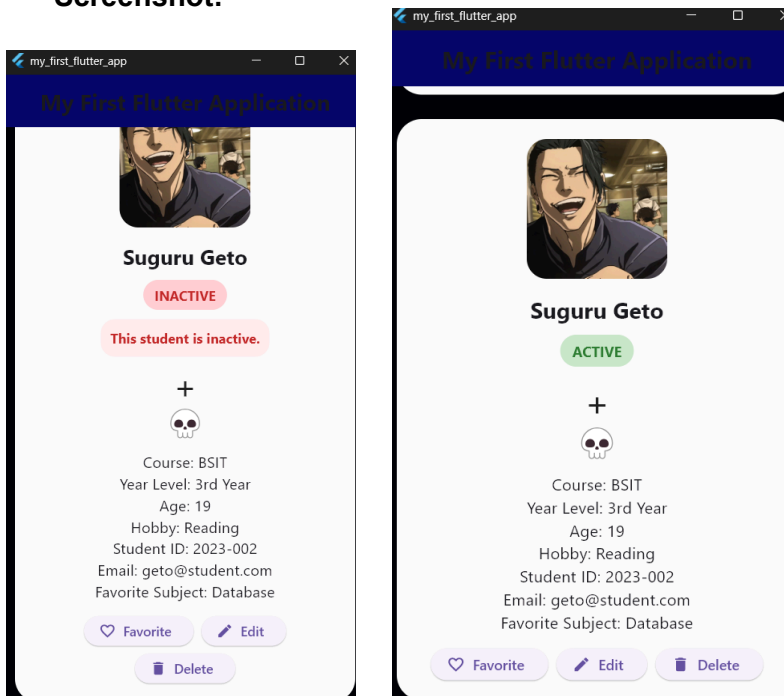
Research ways to conditionally include or exclude widgets from the widget tree.

Test

When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:



```
profile.active == false
? Container(
  padding: const EdgeInsets.all(10),
  margin: const EdgeInsets.only(
    bottom: 5,
  ), // EdgeInsets.only
  decoration: BoxDecoration(
    color: Colors.red.shade50,
    borderRadius: BorderRadius.circular(15),
  ), // BoxDecoration
  child: Text(
    "This student is inactive.",
    style: TextStyle(
      color: Colors.red.shade800,
      fontWeight: FontWeight.bold,
    ), // TextStyle
  ),
```

🚩 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

"No students found"

Empty-state icon

Button to add a student

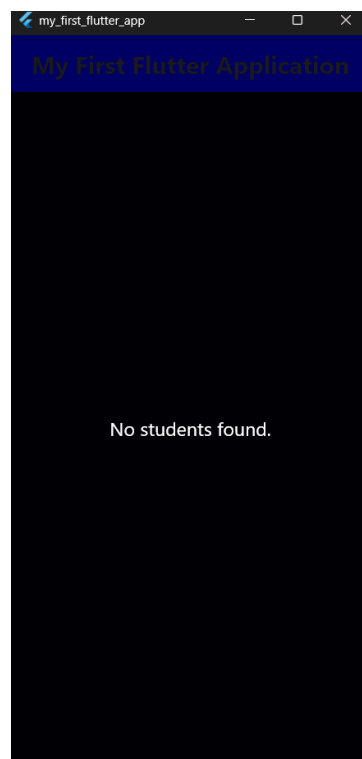
Test

Delete all students in your application (not the code)

Instead of showing a blank screen, your application should display your empty state.

Screenshot:

```
// End of Empty Directory
body: students.isEmpty
  ? const Center(
    child: Text(
      "No students found.",
      style: TextStyle(
        color: Colors.white,
        fontSize: 18,
      ), // TextStyle
    ), // Text
  ) // Center
: ListView.builder(
```



🚩 Flag 4 — Loading the Directory

Objective: Display different UI while student data is being loaded.

Create a loading state for your Student Directory.

Research how Flutter can conditionally display a loading indicator.

Examples:

`CircularProgressIndicator`

Loading message

Loading placeholder

Test

When the application is in a loading state, the student list should not be displayed as the main content.

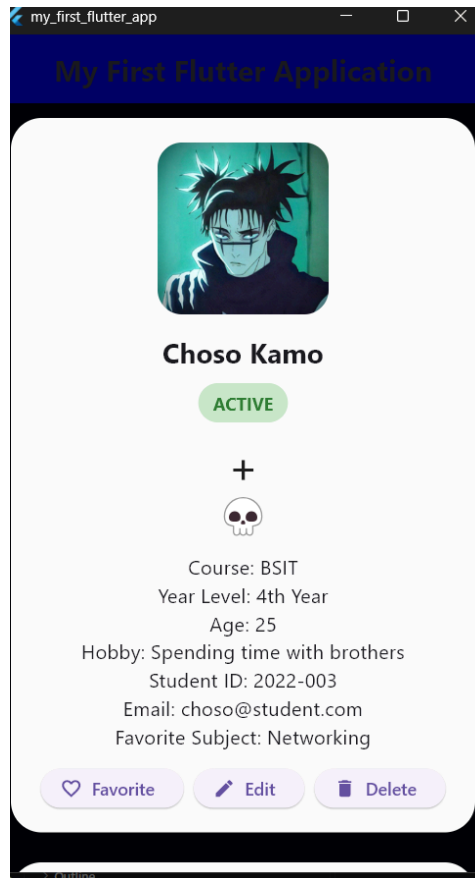
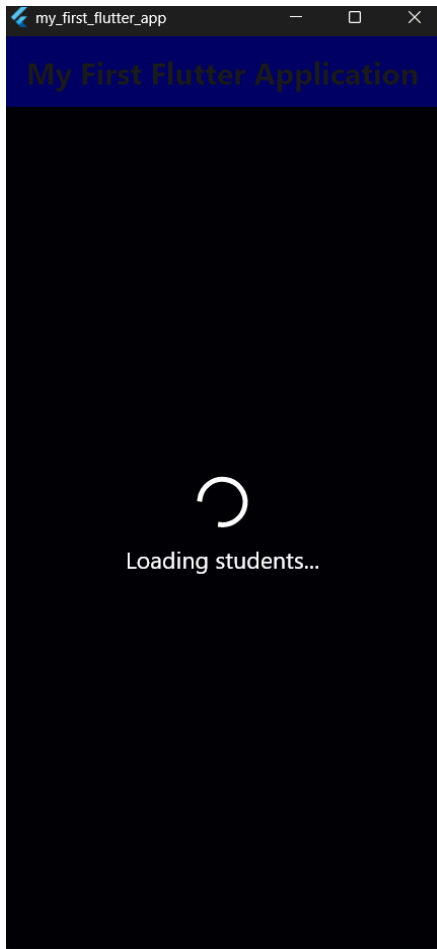
Once loading is complete, the Student Directory should appear.

Screenshot:

```
bool isLoading = true;
```

```
Future.delayed(const Duration(seconds: 3), () {  
  if (mounted) {  
    setState(() {  
      isLoading = false;  
    });  
  }  
}); // Future.delayed  
}
```

```
body: isLoading  
  ? const Center(  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        CircularProgressIndicator(  
          color: Colors.white,  
        ), // CircularProgressIndicator  
        SizedBox(height: 15),  
        Text(  
          "Loading students...",
```



🚩 Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

Loading

Students available

No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading

→ Loading UI

Students available

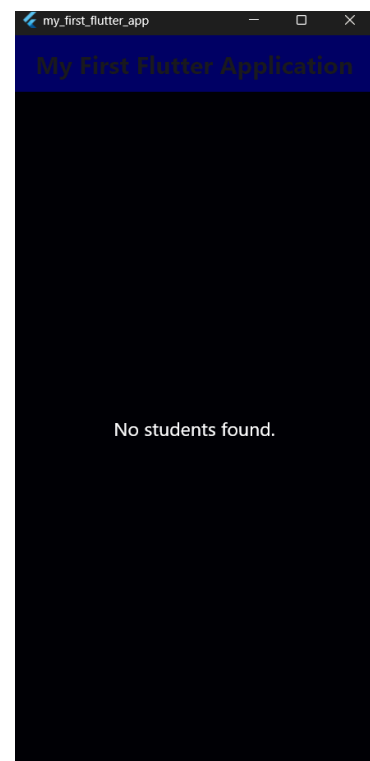
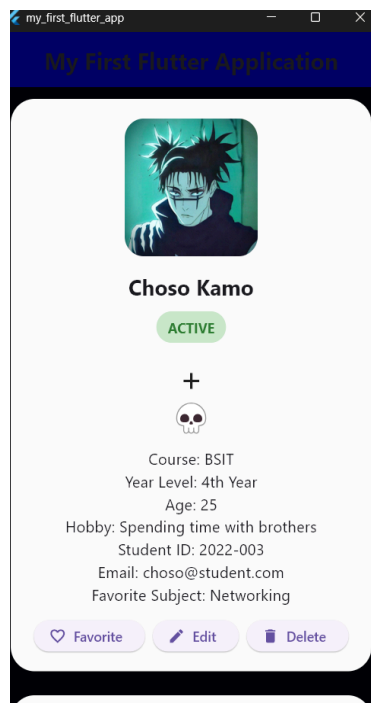
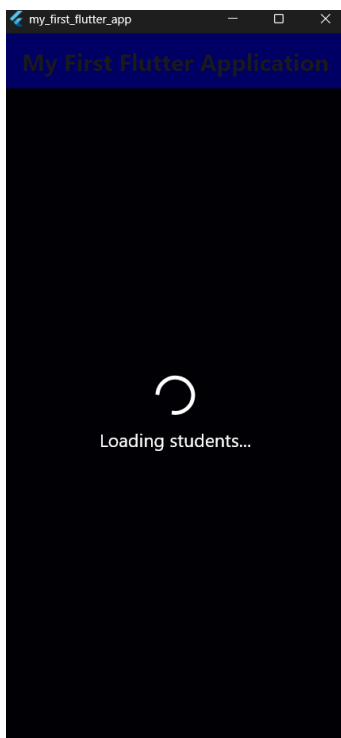
→ Student List

Empty list

→ Empty State

Only the appropriate UI should appear at a time.

Screenshot:



Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

Favorite

Delete

Edit

Active/Inactive

Show/Hide

Empty state

Loading state

Test Scenario

Favorite a student.

Change or display their Active/Inactive state.

Delete students until the directory becomes empty.

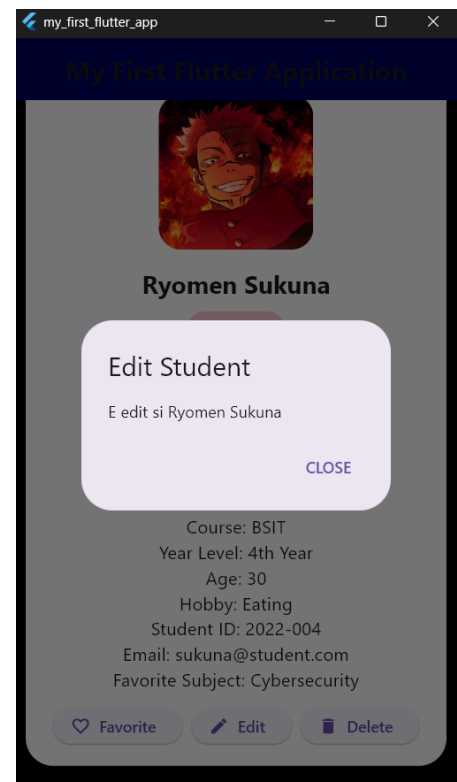
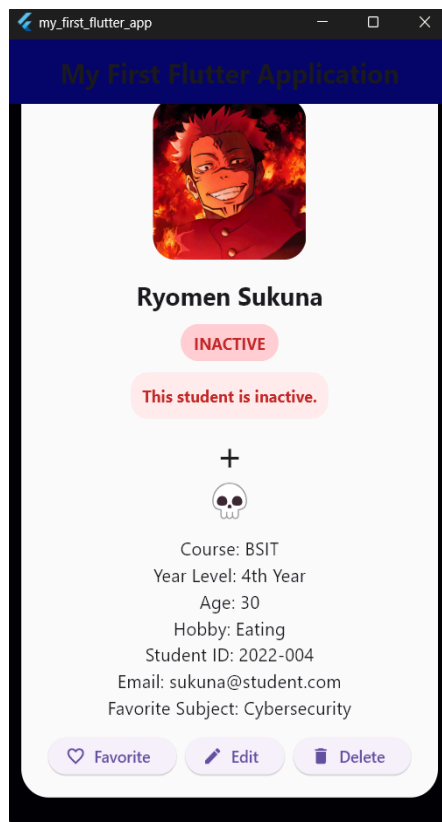
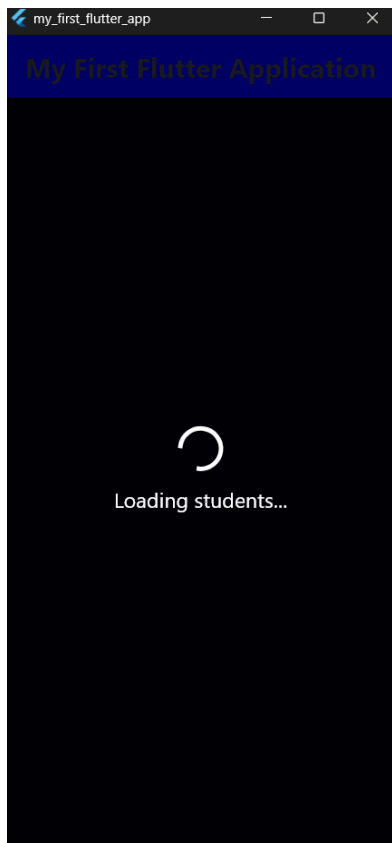
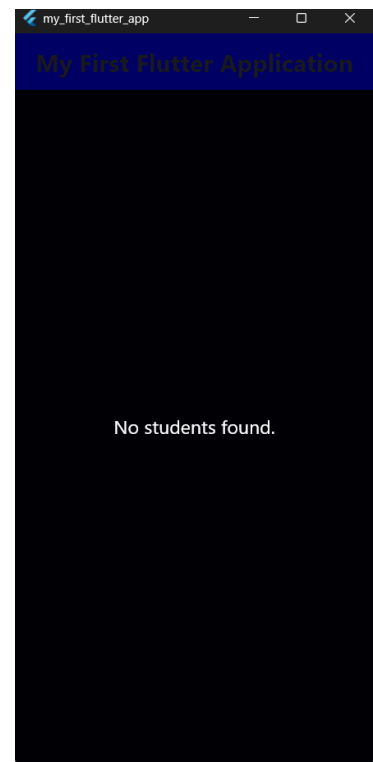
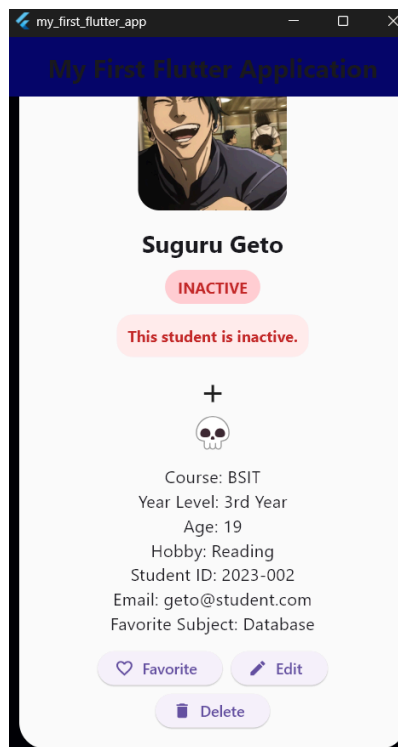
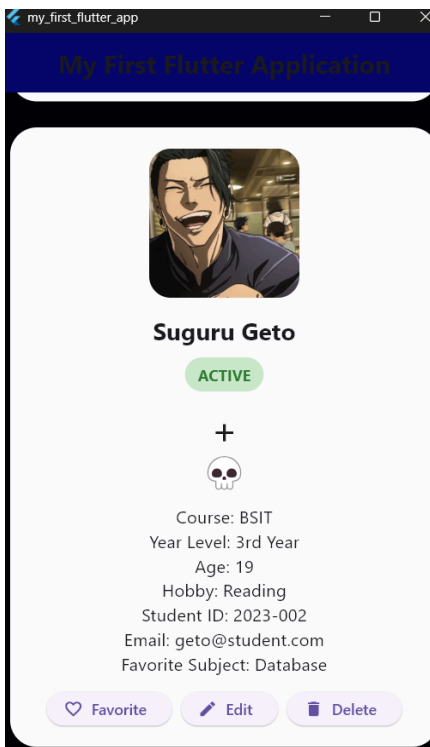
Verify the Empty State appears.

Reload or restore students.

Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

Screenshot:



Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

Use a proper Git commit message.

Branch:

`feature/conditional-ui`

Screenshot:

Successful commit

Current branch

Terminal showing the commit

MINGW64:/c/IT314AB_Caballero

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)

\$ git status

On branch feature/conditional-ui

Your branch is up to date with 'origin/feature/conditional-ui'.

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: FlutterActivities/my_first_flutter_app/lib/main.dart

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)

\$ git commit -m "feat: Submit CTF 6"

[feature/conditional-ui 1f13817] feat: Submit CTF 6

1 file changed, 298 insertions(+), 198 deletions(-)

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)

\$ git push

Enumerating objects: 11, done.

Counting objects: 100% (11/11), done.

Delta compression using up to 8 threads

Compressing objects: 100% (4/4), done.

Writing objects: 100% (6/6), 1.84 KiB | 943.00 KiB/s, done.

Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (3/3), completed with 3 local objects.

To https://github.com/mykylicaballero/IT314AB_Caballero.git

9125251..1f13817 feature/conditional-ui -> feature/conditional-ui

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/conditional-ui)

\$ |